



AQ7.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

If $\|a\| = 3$, $\|b\| = 2\sqrt{2}$, and the angle between a and b is 45° , then the value of $a \cdot b$ is

√

and angle between a and b is 45° , then the value of $a \cdot b$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

1 point

Given that $\|a\| = 3$ and $\|b\| = 5$ and $a \cdot b = 7.5$, for two vectors a and b in R^n . Find the angle (in degrees) between the two vectors a and b . Note:

$\cos 60^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$, $\cos 60^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$

√

/2.

☐

30°

☐

60°

☐

120°

☐

150°

No, the answer is incorrect.

Score: 0

Accepted Answers:

60°

1 point

Consider two vectors $a = (3, 4, -1)$, $b = (2, -1, 2)$ in R^3 . Choose the correct options.

☐

Length/Norm of a is $\sqrt{26}$

√

and that of b is 3.

☐

Length/Norm of a is 5 and that of b is 3.

☐

Length/Norm of a is $\sqrt{24}$

√

and that of b is $\sqrt{7}$

√

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☐

Angle between the two vectors is 90 degrees.





Angle between the two vectors is approximately 00 degrees.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Length/Norm of a is $\sqrt{26}$ 26



and that of b is 33.

Angle between the two vectors is 9090 degrees.

1 point

If the dot product of two vectors is zero, then choose the correct option.

- ☐ The two vectors are perpendicular to each other.
- ☐ The two vectors are parallel to each other.
- ☐ The two vectors are exactly the same.
- ☐ The length of the two vectors must be the same.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The two vectors are perpendicular to each other.

Level 2

If $a = (6, -1, 3)$, $b = (4, c, 2)$ where $a, b \in \mathbb{R}^3$ and a and b are perpendicular to each other, then find the value of c ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 30

1 point

1 point

Consider two vectors $a = (1, 2)$, $b = (2, 2)$ and θ is the angle between them. The sum of the two vectors is given by $c = a + b$. Choose the correct options.



Length/Norm of c is 55.



Length/Norm of c is 2525.



Length/Norm of c is 44.



$\cos \theta = \frac{3}{\sqrt{10}}$ $\cos \theta = \frac{1}{10}$



3.



$\cos \theta = \frac{3}{\sqrt{5}}$ $\cos \theta = \frac{5}{5}$



3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Length/Norm of c is 55.

$$\cos \theta = \frac{3}{\sqrt{10}} \cos \theta = 10$$

√

3.

1 point

Consider 3 vectors a, b, c in R^3 and a scalar λ in R . Choose the set of correct options.

Note: $(\cdot)(\cdot)$ represents the dot product.

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$$\lambda(a \cdot b) = (\lambda a) \cdot b \quad \lambda(a \cdot b) = (\lambda a) \cdot b$$

☐

$$\lambda(a \cdot b) = (\lambda b) \cdot a \quad \lambda(a \cdot b) = (\lambda b) \cdot a$$

☐

$$\lambda(a \cdot b) = (\lambda a) \cdot (\lambda b) \quad \lambda(a \cdot b) = (\lambda a) \cdot (\lambda b)$$

☐

$$(a + c) \cdot b = a \cdot b + c \cdot b \quad (a + c) \cdot b = a \cdot b + c \cdot b$$

☐

$$(a + c) \cdot b = a \cdot c + c \cdot b \quad (a + c) \cdot b = a \cdot c + c \cdot b$$

☐

$$a \cdot a = 0, b \cdot b = 0 \text{ if and only if } a, b \text{ are null vectors.}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\lambda(a \cdot b) = (\lambda a) \cdot b \quad \lambda(a \cdot b) = (\lambda a) \cdot b$$

$$\lambda(a \cdot b) = (\lambda b) \cdot a \quad \lambda(a \cdot b) = (\lambda b) \cdot a$$

$$(a + c) \cdot b = a \cdot b + c \cdot b \quad (a + c) \cdot b = a \cdot b + c \cdot b$$

$$a \cdot a = 0, b \cdot b = 0 \text{ if and only if } a, b \text{ are null vectors.}$$

1 point

Consider two vectors $a = (1, 3, 5, 7, 9)$ and $b = (2, 4, 6, 8, 10)$. Choose the set of correct options.

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The length of b is more than a .

☐

The length of a is more than b .

☐

The length of $(a - b)$ is $\sqrt{5}$.

☐

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☐

The length of $(a - b)$ is $\sqrt{50}$.

☐

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No, the answer is incorrect.

Score: 0

Accepted Answers:

The length of b is more than a .

The length of $(a - b)$ is $\sqrt{5}$.

☐

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Check Answers

Your score is: 0/8